DenialDefender

Autonomous Insurance Appeals
for Hospital Revenue Cycle Teams

We do not charge if we do not recover.

Project Brief: Version 3.0
Event: AMD Developer Hackathon 2026
Track: Vision and Multimodal AI
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Date: May 10, 2026

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1. Executive Summary

United States hospitals and physician practices leave hundreds of billions of dollars in recoverable revenue on the table every year because writing insurance appeal letters is too slow and too expensive to scale. The economics of denials are well understood inside healthcare and almost invisible outside it. Insurers know that nearly every patient and most providers will give up before mounting a full appeal, even though appeals, when filed, succeed at remarkably high rates.

DenialDefender is the autonomous appeals layer for hospital revenue cycle management. A clinic or RCM service uploads a denial letter; the system ingests the denial, the patient chart, the payer medical policy, the relevant clinical literature, and the bank of past successful appeals at that payer, then drafts a submission-ready appeal packet (medical necessity letter, code corrections, supporting evidence, payer-specific phrasing) in minutes instead of days.

DenialDefender sells to hospital revenue cycle teams and to the third-party RCM services they outsource billing to (R1 RCM, Waystar, Ensemble, Conifer, and the long tail). Pricing is contingency-based: a percentage of recovered revenue. Customers carry zero downside, and incentives are perfectly aligned with theirs.

Metric	Value	Source
In-network claims denied (ACA marketplace)	19%	KFF, 2024 ACA marketplace data
Denials ever appealed (patient-side)	Less than 1%	KFF, 2024
Appeal success rate when filed	44 to 82%	HFMA / Premier 2024; KFF
Total denied claim value annually (US)	Approx. $262 billion	KFF + Premier hospital data
Stranded recoverable revenue (never appealed)	Approx. $110 billion	Derived from KFF + Premier

This brief makes the case for why this problem is real, why it is unsolved, why now is the moment, and why AMD Instinct MI300X is not just a competent choice for the workload but the architectural enabler of the product economics.

2. The Problem

Every claim a US healthcare provider submits passes through a payer review process that can deny it for medical, administrative, coding, prior authorization, or unspecified reasons. Denial rates have climbed every year of the past decade, the cost to rework each denial has climbed with it, and the gap between denials filed and denials appealed has widened into one of the largest unautomated workflows in the US economy.

2.1 The denial rate is rising, not falling

KFF analysis of CMS Transparency in Coverage data shows a 19% in-network denial rate and 37% out-of-network denial rate on ACA marketplace plans in 2024, with significant variation across insurers. Some major payers denied more than a third of claims.

Provider-reported numbers are even more alarming. The Experian Health 2025 State of Claims survey of 250 revenue cycle leaders found that 41% of providers report denial rates above 10%, up from 30% in 2022. Among hospital systems specifically, Premier Inc. reported an average initial denial rate of 15% across surveyed hospitals.

> *Health claims are still stuck in a cycle of denials, delays and data errors. 41% of survey respondents said that at least one in ten claims is denied. That is a lot of rework and lost revenue that providers were counting on.*
>
> *Experian Health, State of Claims Report 2025*

2.2 Almost no one appeals, even though appeals work

This is the fact that breaks the economics open. KFF reports that fewer than 1% of denied ACA marketplace claims are ever appealed by patients. On the provider side, only roughly 12% of Medicare Advantage prior auth denials are appealed, and just 6.4% in fee-for-service Medicare.

Yet when appeals are filed, they win at extraordinary rates:

- **Medicare Advantage appeals:** 81.7% of appealed denials overturned, full or partial (HFMA / Premier 2024).
- **Commercial payer appeals:** 54.3% overturned.
- **ACA marketplace internal appeals:** 44% upheld in patient favor; an additional 27% overturned at external review.
- **State-level analyses:** 60 to 80% overturn rates in several states.

Translation: when providers actually file the appeal, they get paid more often than not. The reason they do not file is simple. Appeals are slow, expensive, and do not scale.

2.3 The cost to rework a denial is breaking RCM teams

Industry estimates put the cost to rework a single denied claim between $25 and $118, with the most-cited figure landing at $43 to $48 per denial. Multiplied across roughly 600 million claims denied annually in the US, that is over $19 billion in pure administrative waste before any clinical decision is even revisited.

AKASA revenue cycle management surveys consistently rank denial management as the single most time-consuming RCM task. Industry consolidation around RCM service companies (R1, Waystar, Ensemble, Conifer) reflects providers inability to staff this work in-house at scale. Even those services still run offshore appeal-writing teams as their core operational model.

2.4 Why the gap exists

Writing a competent appeal letter requires reading and synthesizing five distinct document categories: the denial letter itself, the patient full chart, the payer specific medical policy bulletin for that procedure, the relevant clinical literature establishing medical necessity, and the bank of past appeals (what has worked at this payer, in this specialty, for this code, this year).

A skilled human takes 45 to 90 minutes per appeal. The math on $43 per rework only works because most denials never get appealed at all.

3. The Market

US healthcare is a $5 trillion annual market, of which roughly $1.5 trillion flows through claim and reimbursement workflows. That flow is where DenialDefender lives. The buyer is not the patient and not the doctor. The buyer is the financial operations team that gets measured on recovery rate.

3.1 Sizing the opportunity

Roughly 3.6 billion medical claims are processed in the US each year. Applying conservative payer-mix-weighted denial rates yields approximately 600 million denied claims annually. Premier Inc. 2024 hospital survey put the average denied claim value at slightly over $5,000. Multiplying gives an initially-denied claim base in the range of $250 to $300 billion per year.

Industry estimates place 60 to 65% of denials as recoverable on appeal, but only roughly 35% of denials are ever appealed in any form. The remaining recoverable denials, never appealed at all, represent the structural waste DenialDefender targets.

Layer	Calculation	Annual value
Total denied claims (US, all payers)	Approx. 600M claims at approx. $5K average	Approx. $262 billion
Recoverable share	65% of denied	Approx. $170 billion
Currently appealed	35% of recoverable filed	Approx. $60 billion (filed)
Stranded recoverable revenue	65% of recoverable never filed	Approx. $110 billion
Conservative SOM (5%, year-3 target)	Aspirational, illustrative	Approx. $5.5 billion

These numbers are illustrative, not aspirational. They exclude commercial-plan claims (where transparency data is incomplete), employer-sponsored plans, and international markets. The MGMA estimates that two-thirds of denials are recoverable when practices have the right systems; the gap between recoverable and recovered is exactly the wedge.

3.2 The buyer landscape

The US has roughly 6,100 hospitals and around 230,000 physician practices. The high-leverage sales channel is not direct to either of those. It runs through the RCM service companies that already process billions of claims on their behalf:

- **R1 RCM:** approximately $2.5B annual revenue, processes claims for 94 of the largest US health systems.
- **Waystar:** public, approximately $6B market cap as of 2026, serves 1M+ providers.

- **Ensemble Health Partners:** private, $7B+ valuation.
- **Optum / Change Healthcare:** UnitedHealth-owned, dominant market position.
- **Long tail:** Conifer Health Solutions, athenahealth, AdvancedMD, Tebra, plus mid-market and specialty-vertical players.

These companies sell on recovery rate uplift. Every additional point of recovery is direct margin. They are structurally the right initial customers for a contingency-priced AI appeal layer.

3.3 Tailwinds

- **CMS Interoperability and Prior Authorization Final Rule.** Finalized in early 2024, payers must expose prior authorization decisions and denial data via FHIR APIs by January 2027. This is the data layer DenialDefender needs and that no incumbent has been engineered around.
- **Public sentiment shift.** Following the December 2024 killing of UnitedHealthcare CEO Brian Thompson, denial practices have entered mainstream political discourse for the first time in a generation. A January 2026 KFF poll shows 66% of insured adults consider denials a major problem. Bipartisan congressional pressure is real and rising.
- **Payer voluntary commitments.** In June 2025, major insurers publicly pledged to fix the broken prior authorization system, but follow-through has been limited. The political cost of inaction is now meaningful.
- **LLM capability threshold crossed.** Long-context regulated-text reasoning at the level appeals require was not feasible at production quality before late 2024. The window opened recently.

4. Product Overview: DenialDefender v1 MVP

DenialDefender is an AI system that drafts insurance appeal letters at scale. The user (a billing specialist or RCM analyst) uploads a denial letter and authorizes access to the relevant patient record. The system reads everything required to write a competent appeal and produces a submission-ready packet for human review.

4.1 User journey, end to end

1. **Upload.** Drag and drop the denial letter (PDF, DOCX, or TXT). Scanned faxes are accepted as PDF. The system extracts denial reason, denial code, payer, plan, and patient identifiers.
2. **Authorize chart access.** The system loads the relevant clinical record: physician notes, lab results, imaging reports, prior authorization history. In the v1 MVP this comes from the synthetic case bank described in Section 6.
3. **Review the draft.** Within 60 to 90 seconds the system produces a complete appeal packet (medical necessity letter with payer-specific framing, supporting clinical literature with citations, code corrections where applicable, and a confidence score).
4. **Approve and export.** A human reviewer signs off; the packet is exported as TXT or PDF, or copied to the clipboard for downstream submission via fax, payer portal, or API.
5. **Track and learn.** Outcomes feed back into the system. The model learns what wins at each payer, in each specialty, for each denial code.

4.2 What it is not

Boundaries matter, especially in healthcare AI. DenialDefender does not diagnose, does not prescribe, does not auto-submit appeals without human review, does not replace the clinician medical judgment, and does not make coverage determinations. It produces drafts for trained humans to evaluate and submit. This positioning is identical to how Hippocratic AI and other regulated-AI healthcare players have established trust with payers and providers.

5. Core Product Features

The v1 MVP ships with a tightly scoped feature set focused on the single workflow that matters most for hospital revenue cycle teams: ingest a denial, generate a defensible appeal, route it through a human reviewer.

5.1 File upload and ingestion

DenialDefender accepts denial documentation in three formats covering essentially every real-world workflow currently in production at hospital RCM teams:

- **PDF.** Native digital PDFs (text-layer extraction), scanned PDFs, and faxed PDFs. Scanned and faxed PDFs are routed through the vision-language model for OCR and structured extraction.
- **DOCX (Microsoft Word).** Direct text extraction from Word documents. This format is common when payers communicate denial details by email attachment or when RCM teams maintain structured denial logs.
- **TXT (plain text).** For copy-pasted denial content, EHR exports, and clean payer portal extractions.

The frontend exposes a drag-and-drop zone alongside a standard file picker. Multiple files per denial (denial letter plus EOB plus supporting correspondence) can be uploaded in a single session and are loaded into the same context window.

5.2 Human-in-the-loop review dashboard

The review dashboard is the heart of the product for hospital RCM users. Once a draft is generated, the billing specialist sees a side-by-side view:

- **Left pane:** the original denial letter with extracted fields highlighted (denial code, denial reason, payer, plan, denied service, dates).
- **Right pane:** the generated appeal letter, fully editable as rich text, with inline citations linked back to the source evidence (lab values, policy clauses, literature).
- **Top bar:** confidence score (calibrated probability of overturn), denial category tag, and payer style tag.
- **Action row:** Approve, Edit, Regenerate, Reject. Approving an appeal records the human approval event and unlocks the export controls.
- **Export controls:** Download as TXT, Download as PDF, Copy to Clipboard. All three are first-class options because hospital workflows vary widely on the submission step.

The pitch to RCM leaders is not replace your team. It is give your team 10x output capacity. An appeals specialist who can review and submit 60 high-quality appeals per week instead of 8 is more productive, not displaced. The dashboard is engineered around that pitch.

5.3 Confidence calibration and triage

Every generated appeal carries a calibrated probability of overturn at this payer, given features of the denial and the evidence assembled. This allows the RCM team to triage. High-confidence appeals (the easy wins) clear the queue first. Low-confidence appeals route to senior reviewers. Borderline appeals can be deferred or batched. The calibration is generated by a lightweight classifier trained on appeal features (denial code, payer, specialty, clinical evidence strength) and retrained as outcomes accumulate.

5.4 Multimodal denial intake

Denials arrive in messy formats. The intake layer handles all of them: digital PDFs through text-layer extraction, scanned and faxed letters through Qwen2.5-VL-7B vision-language interpretation, and Word and TXT formats through direct parsing. Handwritten chart annotations (still common in older facilities) are interpreted by the vision model when present in supporting clinical documentation.

5.5 Outcome tracking and learning loop

Every appeal generates structured outcome data. The reviewer records whether the appeal was overturned, upheld, or partially overturned, along with the timeline. Per-payer overturn rates feed back into model selection, retrieval strategy, and prompt engineering. The architecture is in place at v1; full closed-loop retraining is post-pilot.

6. Data Strategy: 30 Synthetic Patient Cases

DenialDefender architecture must be proven on data that is realistic enough to behave like production while remaining strictly free of protected health information. The v1 MVP ships with a foundational dataset of 30 comprehensive synthetic patient cases, generated from open-licensed clinical data sources (Synthea), augmented with synthetic payer correspondence, and reviewed for clinical plausibility. The 30-case bank is the proof harness for the architecture, the demo backbone, and the evaluation set for confidence calibration.

6.1 Why 30 cases, not 3 and not 300

Three cases is anecdote. Three hundred is a research project. Thirty cases is the right size for a 4-day hackathon build that needs to demonstrate generalization across payers, specialties, and denial categories without becoming a data-engineering exercise. The 30 cases span 5 payers and 8 clinical specialties, with a mix of denial codes (medical necessity, prior authorization, coding, eligibility, missing documentation) representative of the real-world distribution reported by Premier and Experian Health.

6.2 What each case contains

Each of the 30 cases is a complete patient record assembled to match the structure that real hospital RCM teams pull from EHR systems when preparing an appeal. Every case includes the following components:

- **Patient identification and demographics.** Synthetic name, date of birth, contact information, gender, ethnicity, and self-reported language preference.
- **Medical history.** Past and present illnesses, prior surgeries, family medical history, social history, and relevant lifestyle factors.
- **Clinical notes and data.** Progress notes, physician orders, consultation notes, discharge summaries, and care plan entries spanning the encounter timeline.
- **Medications and allergies.** Current and past medications with dosages, administration routes, prescribers, and a complete allergy list with reaction severity.
- **Diagnostic reports.** Laboratory results, radiology reports (X-ray, CT, MRI, ultrasound), and pathology reports with reference ranges and clinical interpretations.
- **Administrative data.** Insurance information (payer, plan type, member ID, group number), billing codes (ICD-10, CPT, HCPCS), prior authorization references, and consent forms.
- **Immunization records.** Dates and types of administered vaccines with lot numbers and administering provider where applicable.

Each case is paired with a synthetic denial letter (matching one of the five payer voices represented in the dataset) and the corresponding payer medical policy bulletin. This pairing is what allows the architecture to be exercised end to end on a single uploaded denial.

6.3 How the dataset powers the demo

When a judge or evaluator drags a denial letter into the demo interface, the system uses the patient identifier on the denial to retrieve the matching synthetic case from the 30-case bank. From that case, the relevant clinical span (typically 30K to 80K tokens) is loaded into the working context, alongside the matching payer policy bulletin, the literature retrieval results, and the past-appeal exemplars. The context is then handed to Qwen3-32B for generation. The Data Hub on the commercial site (Section 10) lets evaluators download a single sample case or the entire 30-case repository to inspect the data structure or generate their own test denials.

6.4 Compliance posture

All 30 cases are synthetic. No protected health information is used in the hackathon build. The generation pipeline relies on Synthea (MITRE Corporation, Apache 2.0), CMS open-data resources, the PubMed Central Open Access Subset for literature, and synthesized payer correspondence based on publicly documented appeal-writing guides. Production deployment, described in Section 11, layers HIPAA compliance and signed Business Associate Agreements onto this foundation.

7. Technical Architecture: RAG over Fine-Tuning

DenialDefender is deliberately not a fine-tuning play. The product runs production-grade base models without weight modification, and instead leverages the massive memory capacity and bandwidth of the AMD Instinct MI300X to execute Retrieval-Augmented Generation (RAG) and long-context prompt engineering. This decision is not a compromise driven by hackathon constraints. It is a strategic choice with material implications for product velocity, regulatory posture, and customer trust.

7.1 Why RAG and not fine-tuning

- **Regulatory transparency.** Fine-tuned weights are opaque. Hospital compliance teams and payer partners reviewing an AI vendor want to know exactly why a given appeal said what it said. RAG produces an auditable trail: every cited fact in the generated appeal traces back to a retrieved document span. Fine-tuning trains the answer into the weights and erases the audit trail.
- **Velocity on payer-specific knowledge.** Payer medical policy bulletins update monthly. Fine-tuning each new policy into the model would require continuous retraining cycles. RAG updates the policy in the retrieval index in seconds. The model improves without retraining.
- **Avoiding training-data contamination claims.** A fine-tuned healthcare AI invites contamination disputes (was this trained on PHI, did the training data include the customer cases). Pure-base-model plus RAG sidesteps the question entirely. The model never sees customer data during training because there is no training.
- **Hardware capacity makes it possible.** Long-context RAG only beats fine-tuning when you can fit a very large working context into memory at full precision and high bandwidth. The MI300X 192 GB capacity and 5.3 TB/s HBM3 bandwidth are what make this trade-off favorable in production. On 80GB-class GPUs, RAG-with-long-context is operationally infeasible at the precisions regulated customers require.

7.2 Standard claims knowledge dynamically loaded into context

DenialDefender maintains a structured retrieval index of the standard claims-processing knowledge that any competent appeals specialist relies on. For each generated appeal, the system dynamically retrieves the relevant subset and injects it into the working context window alongside the patient case and payer policy. The knowledge layer covers four pillars:

Pillar 1: Managed care plan structures

- **HMO.** Network restrictions, gatekeeper referral requirements, in-network-only coverage rules, and the appeal grounds these rules typically support.
- **PPO.** Tiered network reimbursement, out-of-network exception pathways, and balance-billing protections relevant to appeals on out-of-network denials.
- **EPO and POS.** Hybrid plan structures and the specific appeal arguments that succeed against each.

- **Medicare Advantage.** Two-level appeal pathway (plan reconsideration, then independent QIO review), CMS-mandated timelines, and the higher overturn rates that make MA appeals especially worth filing.
- **Medicaid managed care.** State-by-state variation in appeal procedures and external review rights.

Pillar 2: Essential supporting documents

- **Itemized hospital bills.** UB-04 line-item structure, revenue codes, and the documentation practices that support medical necessity arguments.
- **Medicare Part A and Part B forms.** Claim form requirements (CMS-1450 institutional, CMS-1500 professional), modifier conventions, and place-of-service code implications.
- **Explanation of Benefits (EOB) parsing.** Field-level interpretation of denial reasons disclosed on the EOB.
- **Letters of medical necessity.** Standard structure (clinical justification, peer-reviewed support, treatment alternatives considered, expected outcome) and the rhetorical patterns that succeed at each major payer.
- **Prior authorization records.** Pre-service authorization numbers, approval scopes, and the gap between authorized service and billed service that frequently triggers denials.

Pillar 3: Key review criteria

- **Medical necessity.** The dominant denial category and the dominant appeal-success category. The system retrieves payer-specific medical necessity definitions, applicable evidence-based guidelines (NCCN, ACC/AHA, USPSTF), and the clinical literature that establishes standard of care for the disputed procedure.
- **Coding accuracy.** ICD-10 diagnosis codes, CPT and HCPCS procedure codes, and the cross-walks between them. The system flags coding mismatches (a CPT code that does not align with the ICD-10 diagnosis) as candidates for code-correction appeals rather than medical-necessity appeals.
- **Modifier usage.** Modifier 25 (significant separately identifiable E and M service), Modifier 59 (distinct procedural service), and the documentation requirements that justify each.
- **Bundling and unbundling.** NCCI edits and the documentation patterns that support unbundling exceptions.

Pillar 4: Common denial reasons (and how to argue each)

- **Missing documentation.** The appeal supplies the missing documentation and references the specific policy clause that requires it.
- **No prior authorization.** The appeal either supplies the prior auth number, demonstrates that the service was authorized under a related code, or argues that the urgency of the clinical situation exempted it from prior auth (urgent and emergent care exceptions).
- **Eligibility issues.** The appeal demonstrates patient coverage on the date of service through enrollment records.

- **Non-covered service.** The appeal argues that the service falls within a covered category as defined by the payer policy, citing the specific policy clause.
- **Medical necessity not established.** The appeal supplies clinical evidence (lab values, imaging findings, prior treatment failures) that establishes medical necessity per the payer specific definition.
- **Coding errors.** The appeal supplies the corrected codes and the supporting documentation for the correction.
- **Timely filing.** The appeal demonstrates compliance with the payer filing deadline through the original submission timestamp.
- **Duplicate claim.** The appeal demonstrates that the claim is for a distinct service through service date and procedure detail.

7.3 Long-context prompt engineering

Generation runs on Qwen3-32B in a single forward pass over a long working context. For the hackathon demo this context is bounded at approximately 64K tokens. The architectural pattern scales to 250K+ tokens under FP8 quantization in production (see Section 8). RAG-based chunking degrades appeal quality measurably because the rhetorical force of a medical necessity letter depends on coherent cross-document reasoning: the patient specific lab values cited in the same paragraph as the payer specific policy clause, supported by the specific paper that establishes standard of care, framed in the rhetorical pattern that has historically won at this payer. Chunking breaks that. The architecture deliberately keeps the entire reasoning context resident in a single context window.

7.4 Why this approach is durable

Because no fine-tuning is involved, the system improves continuously with no retraining cycle. New payer policies, new clinical guidelines, and new exemplar appeals all flow into the retrieval index and become live within seconds. Customers contracting with DenialDefender are not contracting for a frozen snapshot of medical knowledge; they are contracting for a system whose knowledge base updates as the regulatory and clinical environment moves. This is the right shape for a regulated-domain product.

8. Why AMD Instinct MI300X

The economics of DenialDefender depend on producing high-quality, regulated-domain text from very large multi-document working contexts at low marginal cost per appeal. The hardware choice is not an afterthought; it is load-bearing. AMD Instinct MI300X is, in the present analysis, the only commodity-priced GPU that makes the unit economics of contingency-priced appeals work today.

8.1 Workload characterization

A single appeal generation pass requires the model to attend over the following document set:

Input document	Typical size	Token estimate
Denial letter	5 to 20 pages	2K to 8K
Patient chart (relevant span)	50 to 500 pages	20K to 200K
Payer Medical Policy Bulletin	10 to 50 pages	5K to 25K
Supporting clinical literature (5 to 15 papers)	200 to 800 pages	80K to 320K
Bank of past successful appeals at this payer	10 to 50 docs	20K to 100K
TOTAL working context per appeal	All of the above	Approx. 130K to 650K

For the hackathon demo, working contexts are bounded at 64K tokens, sufficient to demonstrate the architectural pattern while keeping inference latency under 90 seconds end to end. The full 250K to 650K production envelope is enabled by the same hardware via FP8 quantization paths discussed in Section 8.6.

8.2 MI300X capacity advantage

The MI300X carries 192 GB of HBM3 on a single device. The relevant comparison is whether a complete inference stack (language model plus vision model plus working KV cache for long contexts) fits on a single device:

GPU	Memory	Bandwidth	Fits Qwen3-32B + Qwen2.5-VL-7B + 64K KV (approx. 129 GB)?
AMD Instinct MI300X	192 GB HBM3	5.3 TB/s	Yes. Single device, approx. 63 GB headroom.
NVIDIA H200 SXM	141 GB HBM3e	4.8 TB/s	Tight. No operational headroom.
NVIDIA H100 SXM	80 GB HBM3	3.35 TB/s	No. Requires 2+ GPUs and tensor-

GPU	Memory	Bandwidth	Fits Qwen3-32B + Qwen2.5-VL-7B + 64K KV (approx. 129 GB)?
			parallel sharding.
NVIDIA A100 80GB	80 GB HBM2e	2.0 TB/s	No. Same as H100 plus bandwidth-limited.

Co-residency on a single device is not a benchmarking nicety. It eliminates inter-device traffic between vision and language stages of the pipeline, which is the dominant latency cost in multimodal RAG systems. On NVIDIA-equivalent setups, the same workload requires either multi-GPU sharding (which doubles the per-appeal compute cost) or aggressive INT4 quantization (which regulated-healthcare customers reject for clinical text).

8.3 Bandwidth advantage on memory-bound inference

Long-context decoder inference is memory-bandwidth-bound, not compute-bound. The MI300X 5.3 TB/s of HBM3 bandwidth is approximately 1.6x the H100 SXM5 3.35 TB/s and approximately 1.1x the H200 4.8 TB/s. For workloads where the model spends most of its time streaming KV-cache across the memory hierarchy (precisely the appeal-generation case), this translates directly into lower per-appeal latency, which translates directly into per-appeal compute cost.

8.4 Co-resident multi-model architecture

DenialDefender uses two distinct models in a single appeal generation pipeline:

- **Qwen3-32B (dense, FP16, approx. 64 GB).** Long-context reasoning, evidence synthesis, and appeal letter generation. The dense architecture is deliberate. For grounded long-form writing where every cited fact must be traceable to retrieved evidence, dense models attend more uniformly across context than mixture-of-experts alternatives.
- **Qwen2.5-VL-7B (FP16, approx. 15 GB).** Vision-language model for handwritten chart pages, scanned EOBs, and faxed denial letters where text-layer extraction is insufficient.

On a single MI300X with 192 GB, both models are co-resident in FP16 with split GPU memory allocation (approximately 0.65 / 0.25 utilization split, with approximately 10% headroom). The handoff from vision-extracted denial intake to language-driven appeal generation happens with no inter-device traffic. Total VRAM footprint with 64K-token KV cache is approximately 129 GB of 192 GB available.

8.5 Cost story

Public cloud pricing for MI300X as of May 2026 ranges from $1.99 to $2.35 per GPU-hour. H100 SXM5 on equivalent providers is $4 to $7 per GPU-hour. Combined with the multi-GPU requirement for the same workload on NVIDIA, the per-appeal compute cost on MI300X is structurally 4 to 8x lower than on H100-based infrastructure.

- At target inference latency (60 to 90 seconds end to end), the per-appeal compute cost on MI300X lands at approximately $0.03 per appeal.

- The H100-equivalent setup, accounting for the 2-GPU requirement and longer per-call latency from tensor-parallel sharding, lands at $0.17 to $0.29 per appeal, or 5 to 9x higher.

At a contingency-pricing business model with average recovered claims around $2,500 and a 12% take rate ($300 per successful appeal), the inference cost differential of $0.03 vs $0.20 is the difference between a fat-margin SaaS and a thin-margin services business. The MI300X choice is the unit-economics enabler that makes contingency pricing structurally defensible.

8.6 Software stack and precision strategy

ROCm 7.0 supports PyTorch and Hugging Face natively at the application level, meaning that the engineering team writing the appeal-generation pipeline does not interact with low-level GPU kernels. Migration from a CUDA-based PyTorch project to ROCm is largely a runtime configuration change. vLLM 0.14+ has first-class ROCm support with Day 0 compatibility for the Qwen3 family, and runs Qwen models with prefix caching, paged attention, and continuous batching, which are the three optimizations that matter most for long-context inference economics. Hugging Face Optimum-AMD provides drop-in inference-optimized model loading with ROCm-aware kernels for the Qwen architecture family.

DenialDefender ships at FP16 inference precision. This is a deliberate engineering choice for the regulated-healthcare market, not a hardware constraint. Hospital revenue cycle teams and RCM service buyers explicitly evaluate AI vendors on precision posture during procurement review, and FP16 with no quantization is the table-stakes answer that clears compliance review at R1, Waystar, and the hospital systems in the target buyer set. FP8 in production becomes available later: once payer-specific gold-standard test sets exist (generated from human-reviewed appeal outcomes), FP8 inference on the same MI300X allows running an 80B-parameter model in the same memory budget while halving per-appeal compute cost. The hardware investment is precision-stage-independent. This precision-stage flexibility (FP16 for trust-building with regulated customers, FP8 for production economics once validated) is unique to high-memory accelerators like MI300X. On 80GB-class GPUs, the FP16 stage is operationally infeasible, forcing the precision choice to be made before customer trust is earned.

9. The Pipeline (Per-Appeal Flow)

The pipeline is a five-stage sequence executed on every uploaded denial. Each stage has a clearly defined input, a clearly defined output, and a clearly defined model or service responsible for it.

9.1 Stage 1: Ingestion and parsing

The user drops a PDF, DOCX, or TXT file into the upload zone (or selects through a standard file picker). The ingestion service detects the format and routes accordingly. Native PDFs and DOCX go through a text-extraction path. TXT goes through direct parsing. Scanned and image-based PDFs are routed to Qwen2.5-VL-7B, which extracts the raw text content along with structural cues (header positions, table boundaries, signature blocks). The output of Stage 1 is a normalized denial document representation: extracted text plus a structured envelope of fields (denial code, denial reason, payer, plan, denied service, dates, claim number).

9.2 Stage 2: Context retrieval (RAG)

Using the patient identifier extracted in Stage 1, the system queries the vector database (PostgreSQL with pgvector) to pull the relevant patient history from the 30-case corpus. In parallel, the system queries the standard claims knowledge index to retrieve the relevant managed care guidelines, required documents, and denial-reason argumentation patterns described in Section 7.2. The payer-specific medical policy bulletin is loaded through exact match on payer plus procedure code. Supporting clinical literature is retrieved through pgvector similarity search over the PubMed Central Open Access Subset. The output of Stage 2 is a complete evidence package: patient history, payer policy, clinical literature, past-appeal exemplars, and standard claims rules.

9.3 Stage 3: Synthesis and reasoning

Qwen3-32B receives the denial reason, the patient clinical history, the payer policy, the clinical literature, and the standard claims knowledge in a single working context window (approximately 40K to 60K tokens for the demo, scaling to 250K+ in production). The model performs the core reasoning task: identify the specific payer rationale for the denial, identify the strongest available counter-argument given the patient evidence and clinical guidelines, identify any code corrections that would resolve the denial without a medical-necessity argument, and select the rhetorical pattern that has historically succeeded at this payer.

9.4 Stage 4: Draft generation

Qwen3-32B generates the appeal letter, citing specific lab results, imaging findings, prior treatment outcomes, and policy clauses by exact reference. The generation is constrained: the model can only cite evidence present in the retrieved context. This grounding constraint is the primary defense against hallucinated citations. The output is a complete draft (typically 800 to 1,500 words for a routine appeal, longer for complex medical-necessity cases) along with a structured metadata block (cited sources, confidence score, recommended escalation level).

9.5 Stage 5: Review and export

The draft surfaces in the human-in-the-loop review dashboard described in Section 5.2. Hospital staff review the draft side by side with the original denial, edit as needed, and click Approve. On approval, export controls activate: Download as TXT, Download as PDF, Copy to Clipboard. The submission step itself (fax, payer portal upload, payer API submission) remains a human decision in v1; the architecture supports payer API submission for the post-2027 CMS Interoperability Rule environment.

9.6 Outcome capture loop

Post-submission outcome (overturned, upheld, partial) is recorded by the reviewer in a follow-up step (typically days to weeks later, when the payer responds). Outcome data feeds the per-payer learning loop: retrieval strategy adjustments, prompt template revisions, and confidence-score recalibration. The architecture is built into v1; closed-loop learning on real outcomes activates post-pilot once outcome volumes are sufficient for statistical significance.

10. UI / UX and Deployment Strategy

DenialDefender ships through two coordinated deployments, each engineered for a distinct audience: a commercial-grade marketing and demo site for prospective customers and the broader market, and a technical evaluation surface for hackathon judges and the open-source community. Both deployments are brand-consistent and share the same underlying inference backend. They differ in framing, surface area, and call-to-action.

10.1 Deployment A: Commercial website (trydenialdefender.com)

Built with Next.js 14 (TypeScript, Tailwind CSS) and deployed to Vercel for low-latency global delivery. The site is the first impression for hospital RCM leaders, RCM service company evaluators, and investors. It is structured in three sections, accessible through both top-level navigation and continuous scroll.

Section 1: Landing page

Professional overview of the product. The fold contains the value proposition, a 30-second product video (or animated explainer), and a single primary call to action (Try the Demo). Below the fold the page covers: the denial-economics problem (compressed version of Section 2), the buyer personas and value proposition (compressed version of Section 12), the foundational data details (the 30-case strategy from Section 6), and the high-level technical architecture (the RAG-on-MI300X story from Sections 7 and 8). The page closes with a section on team Sophon and contact information for design partner inquiries.

Section 2: Interactive demo

The functional core of the marketing site. A drag-and-drop file upload zone occupies the center of the screen, with a secondary file-picker button for accessibility. On upload, a progress indicator displays the active pipeline stage (Ingesting, Retrieving context, Generating draft, Finalizing). Within 60 to 90 seconds the result pane displays the generated appeal alongside the original denial. Three export controls are surfaced as primary buttons:

- **Download as TXT.** Plain text export for users who will paste the appeal into payer portal text fields or further-process the output downstream.
- **Download as PDF.** Formatted PDF suitable for fax submission or printing. Includes the medical necessity letter structure, citations, and a signature block.
- **Copy to Clipboard.** Single-click copy of the full appeal text for fast pasting into any destination application.

Below the result pane, an Edit Inline option lets evaluators experience the human-in-the-loop revision flow. Confidence score, denial category, and payer style tag are displayed prominently above the draft.

Section 3: Data Hub

An embedded list and distinct button section that allows users to download the synthetic dataset for independent evaluation. Two download paths are exposed:

- **Download a single sample case.** A single representative synthetic patient case as a ZIP bundle (patient data, denial letter, payer policy, supporting evidence). Useful for quick evaluation without committing to the full repository.
- **Download the full repository (30 cases).** The complete 30-case corpus as a single archive, indexed by payer, specialty, and denial category. Useful for technical evaluators who want to test the architecture against the entire dataset or generate their own denial scenarios from the underlying patient records.

Each case in the Data Hub is documented with a brief synopsis (payer, specialty, denial reason, expected appeal angle) so evaluators can pick cases that match their interest area. The Data Hub also includes the Synthea generation seeds and the source-pointer documentation, supporting full reproducibility.

10.2 Deployment B: Technical evaluation space (Hugging Face / Gradio)

Built as a Gradio application, deployed as a Hugging Face Space within the AMD Developer Hackathon organization. This deployment is optimized for judge review and asynchronous technical evaluation, including by remote judges who will never see the on-site live demo. The commercial landing-page surface is deliberately omitted; the framing is technical and evaluation-first.

The Space presents the same interactive demo workflow (Upload -> Generate -> Review / Export) at the top of the interface, with the export controls (TXT, PDF, Copy) preserved. Below the demo, three collapsible sections (or alternatively rendered as tabs depending on the Gradio theme) expose the technical depth a judge needs to evaluate the project:

- **Project Justification.** Compressed version of the executive summary, the problem framing, and the business case. Covers Sections 1 through 3 of this brief.
- **Data Strategy.** The 30-case dataset structure, the per-case component breakdown, and the synthetic-data compliance posture. Includes inline links to download a sample case or the full corpus, mirroring the Data Hub on the commercial site.
- **Technical Architecture.** The RAG-over-fine-tuning explanation, the standard claims knowledge layer, the MI300X capacity and bandwidth advantages, and the per-appeal pipeline diagram. Covers Sections 7 through 9 of this brief.

The Space is engineered to acquire likes from the Hugging Face community (a hackathon prize criterion). The demo surface is the first thing a visitor sees, the architecture story is one click away, and the visual identity matches the commercial site exactly so a visitor moving between the two surfaces sees a single coherent product.

10.3 Universal metadata and footer

Both deployments share an identical footer block, ensuring brand consistency and clear attribution across the project surface. The metadata exposed in the footer is:

- **Project Title:** DenialDefender
- **Team:** Sophon
- **Core Member:** Jorge Sandoval

- **Open Source:** github.com/jorgesandoval/denialdefender
- **Contact:** contact@jorgesandoval.dev
- **Personal Website:** jorgesandoval.dev
- **Socials (X and LinkedIn):** @jorgesandev

The footer also includes the AMD Developer Hackathon 2026 acknowledgment, the Vision and Multimodal AI Track tag, and a hackathon-event Hugging Face organization link. Brand colors, typography, and spacing rules are shared between deployments through a single design-token configuration to guarantee visual consistency.

11. Tech Stack Summary

11.1 Compute layer

- **Hardware:** AMD Instinct MI300X via AMD Developer Cloud. Hackathon allocation covered by the $100 program credits for AMD AI Developer Program members.
- **Software:** ROCm 7.0 runtime, AMD CDNA 3 architecture-aware kernels.
- **Inference server:** vLLM 0.14+ with ROCm backend, prefix caching, paged attention, continuous batching.
- **Model loading:** Hugging Face Optimum-AMD.

11.2 Models

- **Primary reasoning model:** Qwen3-32B (dense, FP16). Long-context appeal generation, payer-specific style transfer, evidence synthesis.
- **Vision model:** Qwen2.5-VL-7B (FP16). Denial letter OCR, scanned chart page interpretation, structured extraction from faxed payer correspondence.
- **No fine-tuning.** Both models run as released weights. All domain adaptation occurs through RAG and long-context prompt engineering (Section 7).

11.3 Application layer

- **Backend:** FastAPI (Python 3.11), async request orchestration, structured outputs via Pydantic models.
- **Storage:** PostgreSQL 16 with pgvector extension for past-appeal and literature retrieval; encrypted at rest.
- **Frontend (commercial):** Next.js 14, TypeScript, Tailwind CSS, deployed to Vercel.
- **Frontend (HF Space):** Gradio, deployed as a Hugging Face Space within the event organization.
- **Document generation:** Server-side PDF generation for the Download as PDF export.

11.4 Data sources

All data used during the build is synthetic or open-licensed. No protected health information is involved.

- **Synthetic patient records:** Synthea (MITRE Corporation, Apache 2.0).
- **Payer policies:** CMS Medicare Coverage Determinations (public domain) and synthetic commercial-payer policy bulletins.
- **Clinical literature:** PubMed Central Open Access Subset.
- **Denial letters and appeal templates:** Synthesized from publicly available appeal-writing guides and de-identified appeal corpora published in academic literature.

11.5 Production posture (out of hackathon scope)

In production deployment, DenialDefender is HIPAA-compliant with signed BAAs with each customer; SOC 2 Type II is on the year-one roadmap. PHI handling uses tenant-isolated encrypted stores and explicit data residency controls. None of this is built for the hackathon. The hackathon ships the synthetic-data demo and the architecture diagram. The production posture is described here so judges understand the path forward, not because the team claims to have shipped it in 4 days.

12. Buyer Personas and Go-to-Market

Healthcare software lives or dies on buyer selection. The best technology pointed at the wrong buyer becomes a graveyard. DenialDefender targets two buyer segments and explicitly avoids a third.

12.1 Primary buyer 1: RCM service companies

Companies like R1 RCM, Waystar, Ensemble Health Partners, Conifer, and a long tail of mid-market RCM services process billing on behalf of hospitals and physician groups. Their entire P&L is driven by recovery rate uplift. A 1-point improvement in recovery is direct margin to them and a sticky retention story for their customers. These companies already have integration teams, security review processes, and procurement infrastructure that can absorb a vendor like DenialDefender. They are not selling to doctors. They are selling to CFOs, exactly like DenialDefender does.

- **Sales motion:** design partner program with one mid-market RCM service (200 to 500 hospital clients), prove recovery uplift in a 90-day pilot on a defined claim set, expand to enterprise contract.
- **Initial deal size:** $250K to $1M ARR; expansion potential 5 to 10x per logo.

12.2 Primary buyer 2: Hospital systems (200 to 1,000 beds)

Mid-market hospital systems run RCM in-house and lose tens of millions per year in unrecovered denials. The buyer inside the hospital is the VP of Revenue Cycle, who reports to the CFO and whose KPI is net collection rate. They have budget authority for tools that demonstrably improve recovery. The pitch is not replace your team. The pitch is give your team 10x output capacity. An appeals specialist who can review and submit 60 high-quality appeals per week instead of 8 is more productive, not displaced.

- **Deal size:** $50K to $500K ARR per system, scaling with bed count and payer mix.

The roughly 6,100 US hospitals plus the largest physician group practices represent a $1B+ direct serviceable market once channel is established.

12.3 Who DenialDefender explicitly does not sell to

Solo doctors and small clinics (under 5 providers) feel the pain most acutely but are the worst possible customer. They have no procurement process, no IT, no budget for SaaS, and the LTV math does not work. Every healthcare startup graveyard is full of we-will-sell-to-small-clinics pitches. DenialDefender acknowledges this segment exists, routes them to partner products (a freemium tier or a referral to ReachRx-style platforms), and stays focused on the buyers who can pay.

12.4 The wedge

We do not charge if we do not recover. DenialDefender takes a percentage of revenue restored that the customer would otherwise have written off. Zero upfront. Zero downside. We win when the customer wins. This sentence makes the deal close. CFOs cannot say no to a vendor that costs nothing if it does nothing. It also gracefully disqualifies customers whose denial volumes are too small to be worth the engagement, which is exactly the segmentation discipline the GTM motion requires.

13. Business Model

13.1 Pricing: contingency, not seats

DenialDefender charges a percentage of recovered revenue (industry-standard contingency pricing in the 8 to 15% range, depending on segment and volume). RCM service companies negotiate the lower end (8 to 10%) at scale; mid-market hospital systems pay closer to 12 to 15% on a smaller base. The pricing matches how the RCM industry already buys. This is not the SaaS-default model and that is the point. Seat-based pricing creates procurement friction in a buyer that already runs SaaS budget audits quarterly. Contingency pricing turns the conversation from how much does this cost to how much do you find, which is the question RCM leaders want to be asked.

13.2 Unit economics: illustrative

Consider a representative 250-bed community hospital:

Variable	Value
Annual claim volume	Approx. 120,000 claims
Initial denial rate (15%)	Approx. 18,000 denied claims
Average denied claim value	$5,200
Total denied claim value annually	Approx. $93.6M
Currently appealed (35%) and recovered (54%)	Approx. $17.7M recovered
Stranded recoverable revenue (65% never appealed at 60% recoverable)	Approx. $36.5M
DenialDefender uplift target (recover 30% of stranded)	Approx. $11M new recovery
DenialDefender take at 12% contingency	$1.32M ARR per system

Even discounting that example by half for early-stage execution risk gives $600K to $700K ARR per mid-market hospital. Closing 10 hospitals in year one is $6 to $7M ARR. Closing one mid-tier RCM service company with 200 client hospitals is meaningfully more. The unit economics support a Series A in year two on conservative assumptions.

13.3 Defensibility

- **Proprietary outcome data.** Every appeal generated produces a structured win / loss outcome tied to payer, specialty, denial code, and rhetorical strategy. Within 12 months DenialDefender holds the largest dataset of what wins on appeal in the industry, per payer, per code. This is the moat.

- **Per-customer learning loop.** The retrieval layer adapts on the customer actual claim mix, not generic data. Switching costs accumulate. A year of payer-mix-specific exemplar growth on a hospital is not portable to a competitor.
- **Network effects across payers.** What wins at Aetna teaches the system something about what wins at UnitedHealth. Cross-payer pattern transfer compounds across the customer base.
- **CMS data tailwind.** When the Interoperability Rule payer-API requirement takes effect in 2027, customers already onboarded inherit a structured-denial-data advantage immediately.

13.4 Acquisition logic

R1 RCM, Waystar, Optum, athenahealth, and the major hospital RCM platforms all need this capability and are not going to build it from scratch in time. The exit path through strategic acquisition is well-paved by recent comparables: Cohere Health at approximately $589M valuation on the payer side, Harvey AI at approximately $5B in legal AI, Abridge at approximately $5.3B in clinical scribing. A focused vertical AI company with proprietary outcome data and recurring contingency revenue is exactly the asset profile these acquirers pay premium multiples for.

14. Competitive Landscape

Healthcare AI is a crowded space. The right competitive question is not is anyone in this room. They are. The right question is is anyone doing this specific thing for this specific buyer. The answer is no, for reasons that are structural and durable.

14.1 Adjacent but not overlapping

Company	What they do	Why it is not us
Cohere Health (approx. $589M val)	Intelligent prior auth on the payer side	They sell to insurance companies, helping payers make smarter coverage decisions. We sell to providers fighting their decisions. Complementary, not competitive.
Rhyme, Latent Health, Availity	Prior auth submission tools on the provider side	Submission, not appeals. Different workflow, different buyer in the org.
R1 RCM, Waystar, Optum, athenahealth	Full-stack RCM platforms	Rules-based, pre-LLM denial workflows. Potential acquirers, not competitors.
Hippocratic AI ($126M Series C)	Patient-facing voice agents (post-discharge calls, intake)	Different workflow entirely. Their explicit constraint is no clinical decision support.
Abridge, Suki, Nuance DAX	AI medical scribing	Clinical documentation at point of care. Pre-claim, not post-denial.
Harvey AI (approx. $5B val)	Legal document AI for law firms	Wrong vertical and wrong buyer.

14.2 Why no one is in the same lane

The provider-side appeals workflow has been structurally underserved because it required three things to converge that have only converged in the last 18 months: (1) LLMs capable of reliable long-context regulated-text reasoning, (2) commodity GPU infrastructure capable of running those models at per-appeal economics that match contingency pricing, and (3) regulatory data infrastructure (CMS Interoperability Rule) that exposes denial data via APIs. Until 2024, you had at most one of those. By late 2025, you have all three. The window has opened.

14.3 Positioning

DenialDefender is the only provider-side, contingency-priced, AI-native appeal layer designed for RCM service operators. Cohere wins the payer-side market; DenialDefender wins the provider-side. Both can be large companies.

15. Why Now

Four independent lines of force converge in 2026 to make this the right moment:

- **Regulatory.** Payer denial data becomes accessible. The CMS Interoperability and Prior Authorization Final Rule, finalized in early 2024, requires payers to expose prior authorization decisions and denial data via FHIR APIs by January 1, 2027. This is the data infrastructure DenialDefender is engineered around. Incumbents whose denial workflows were built before this rule will have to retrofit; DenialDefender will be native.
- **Political.** Denial practices have entered mainstream discourse. The December 2024 killing of UnitedHealthcare CEO Brian Thompson did not create the underlying anger about denials, but it surfaced it permanently. A January 2026 KFF poll found 66% of insured adults consider delays and denials a major problem. Bipartisan legislation targeting prior authorization transparency has advanced in both houses. Insurers June 2025 voluntary commitments to fix the broken prior authorization system reflect the political cost of continued inaction. Hospitals advocating for fairer adjudication have political cover they did not have two years ago.
- **Technical.** Long-context regulated-text reasoning crossed the threshold. Producing a high-quality, defensible appeal letter requires reading 200K+ tokens of mixed clinical, regulatory, and legal text and synthesizing them into a rhetorically coherent argument. This was infeasible at production quality before late 2024. With Qwen3, recent Llama models, and the broader open-weights long-context generation, the capability is now at the threshold where it beats human throughput at acceptable quality, with humans in the loop for review.
- **Economic.** AMD MI300X makes per-appeal economics viable. Contingency pricing requires per-appeal compute costs in single-digit dollars, ideally well under $2. On NVIDIA H100-based infrastructure with the multi-GPU sharding that long-context regulated-domain inference requires, that math does not close. On MI300X, with 192 GB enabling single-device co-residency of a 32B-class language model and a 7B vision model at full FP16 precision, and 5.3 TB/s bandwidth keeping latency low, it does. This is a 2025 to 2026 unlock.

16. Risks and Mitigations

- **Regulatory and compliance risk.** PHI handling, HIPAA, BAA execution, state-by-state insurance regulation, and emerging AI-in-healthcare regulation are real concerns. Mitigation: explicit human-in-the-loop on every submitted appeal, BAA-first sales motion, no clinical decision-making by the system, alignment with the regulated-AI positioning that Hippocratic AI and similar players have validated with payers.
- **Hallucination risk in regulated text.** An appeal letter that cites a non-existent paper or a misattributed clinical fact is worse than no appeal. Mitigation: every cited reference is grounded in the retrieved evidence corpus and verified against the source before inclusion. The model does not generate citations from internal knowledge. It composes only from retrieved spans. Confidence scoring filters low-confidence drafts before they reach human reviewers.
- **Payer countermeasures.** If DenialDefender succeeds, payers will respond by making denials more sophisticated. This is real and welcomed as a market signal. Mitigation: the per-payer learning loop adapts faster than payer policy review cycles; structurally, the harder a payer makes appeals to write, the more value DenialDefender provides; and Cohere Health traction on the payer side suggests the sophistication arms race is happening regardless of whether DenialDefender participates.
- **Insurance industry lobbying.** The industry has significant political capital and may push for restrictions on AI-generated appeals. Mitigation: the system is explicit that human reviewers approve every appeal. It is an authorship tool, not an autonomous filer. Provider trade associations (AHA, AMA) have aligned interests and significant counter-lobbying capacity.
- **Demo risk at the hackathon itself.** Live demos fail. Mitigation: pre-recorded fallback video, two independent demo paths (live MI300X plus cached HF Space), and a synthetic dataset prepared and tested before SF.

17. Hackathon Scope (4 Days)

This brief describes a multi-year company. The hackathon ships a sharp slice of it. Distinguishing the demo from the company is part of how DenialDefender earns judge trust.

17.1 What ships by Saturday May 10, 12:00 PT

- Working DenialDefender pipeline running on AMD Instinct MI300X via the AMD Developer Cloud.
- Qwen3-32B (FP16) and Qwen2.5-VL-7B (FP16) co-resident on a single MI300X, served via vLLM 0.14+ with ROCm 7.0.
- Total VRAM utilization approximately 129 GB of 192 GB available (approximately 67%), demonstrating the headroom that enables the FP16-to-FP8 production migration path described in Section 8.6.
- Commercial website (Next.js, trydenialdefender.com) with landing page, interactive demo, and Data Hub.
- Hugging Face Space (Gradio) deployed within the AMD Developer Hackathon organization, with the same demo workflow and collapsible technical documentation.
- End-to-end demo: synthetic denial letter (PDF, DOCX, or TXT) to 90-second appeal generation to side-by-side review UI to TXT / PDF / Clipboard export.
- Foundational dataset of 30 comprehensive synthetic patient cases across 5 payers and 8 specialties.
- Open-source GitHub repository at github.com/jorgesandoval/denialdefender with complete reproducibility instructions (qualifies for Build in Public challenge).
- Two technical updates posted to X and LinkedIn during the hackathon, tagging @lablab and @AIatAMD (qualifies for Build in Public challenge).
- This brief, the architecture diagram, and the 3-minute pitch deck.

17.2 What is roadmap, not demo

- Real EHR integrations (Epic, Cerner, Meditech). Production-only.
- Live payer API integrations under the CMS Interoperability Rule. 2027 timeline.
- HIPAA-compliant production deployment with signed BAAs.
- SOC 2 Type II certification.
- Outcome-tracking learning loop closure on real customer outcomes (architected, not running on real outcomes during the hackathon).
- Per-payer fine-tuned LoRA adapters: explicitly out of scope. The architecture is RAG plus long-context, not fine-tuning (see Section 7).

17.3 Targeted prizes

- **Vision and Multimodal AI Track 1st Place:** $2,500.

- **Grand Prize (Best Overall Project):** $5,000.
- **Hugging Face Special Prize 1st Place:** Reachy Mini Wireless plus 6 months HF Pro plus $500 HF Credits, driven by HF Space likes.
- **Build in Public Reward:** Awarded for technical updates and open-source release.
- **Social Engagement Hardware Reward:** AMD Radeon AI PRO R9700 GPU, awarded for outstanding social engagement and project promotion.

18. Project Metadata and Contact

DenialDefender is a Team Sophon submission to the AMD Developer Hackathon 2026, Vision and Multimodal AI Track. The following metadata is shared identically across the commercial website, the Hugging Face Space, the GitHub repository, and the pitch surface.

Field	Value
Project Title	DenialDefender
Tagline	Autonomous insurance appeals for hospital revenue cycle teams
Team	Sophon
Core Member	Jorge Sandoval
Track	Vision and Multimodal AI
Hackathon	AMD Developer Hackathon 2026
Open Source Repository	github.com/jorgesandoval/denialdefender
Commercial Demo	trydenialdefender.com
Hugging Face Space	huggingface.co/spaces/lablab-ai-amd-developer-hackathon/ denialdefender
Contact Email	contact@jorgesandoval.dev
Personal Website	jorgesandoval.dev
X (Twitter)	@jorgesandev
LinkedIn	@jorgesandev

All inquiries from prospective design partners, hospital RCM leaders, RCM service evaluators, AMD Developer Cloud staff, and Hugging Face community members can be directed to contact@jorgesandoval.dev. Response within one business day is the standard.

19. Appendix: Sources

All numerical claims in this brief are sourced to primary or industry-standard secondary publications. Judges fact-checking any figure can find the underlying data via the references below.

19.1 Healthcare denial market data

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19.4 Hackathon and ecosystem references

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